

## SECTION 5

### SIGNAL DESCRIPTION

This section contains brief descriptions of the input and output signals in their functional groups, as shown in Figure 5-1. Each signal is explained in a brief paragraph with reference to other sections that contain more detail about the signal and the related operations.

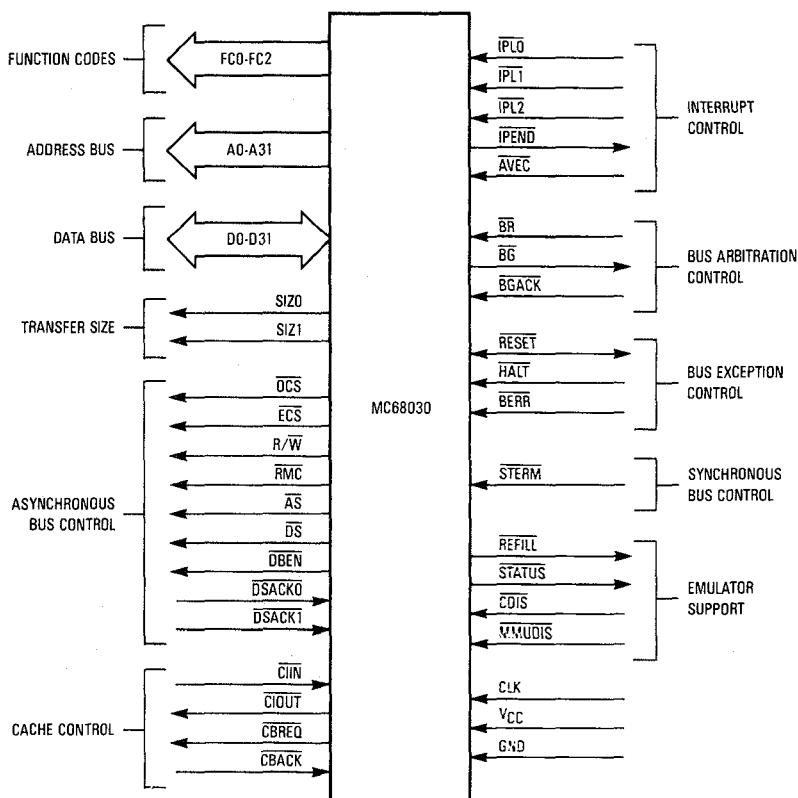


Figure 5-1. Functional Signal Groups

## NOTE

In this section and in the remainder of the manual, **assertion** and **negation** are used to specify forcing a signal to a particular state. In particular, **assertion** and **assert** refer to a signal that is active or true; **negation** and **negate** indicate a signal that is inactive or false. These terms are used independently of the voltage level (high or low) that they represent.

## 5.1 SIGNAL INDEX

The input and output signals for the MC68030 are listed in Table 5-1. Both the names and mnemonics are shown along with brief signal descriptions. For more detail on each signal, refer to the paragraph in this section named for the signal and the reference in that paragraph to a description of the related operations.

Guaranteed timing specifications for the signals listed in Table 5-1 can be found in M68030EC/D, *MC68030 Electrical Specifications*.

Table 5-1. Signal Index (Sheet 1 of 2)

| Signal Name             | Mnemonic                 | Function                                                                                                                                                       |
|-------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Function Codes          | FC0-FC2                  | 3-bit function code used to identify the address space of each bus cycle.                                                                                      |
| Address Bus             | A0-A31                   | 32-bit address bus.                                                                                                                                            |
| Data Bus                | D0-D31                   | 32-bit data bus used to transfer 8, 16, 24, or 32 bits of data per bus cycle.                                                                                  |
| Size                    | SIZ0/SIZ1                | Indicates the number of bytes remaining to be transferred for this cycle. These signals, together with A0 and A1, define the active sections of the data bus.  |
| Operand Cycle Start     | $\overline{\text{OCS}}$  | Identical operation to that of $\overline{\text{ECS}}$ except that $\overline{\text{OCS}}$ is asserted only during the first bus cycle of an operand transfer. |
| External Cycle Start    | $\overline{\text{ECS}}$  | Provides an indication that a bus cycle is beginning.                                                                                                          |
| Read/Write              | $\overline{\text{R/W}}$  | Defines the bus transfer as a processor read or write.                                                                                                         |
| Read-Modify-Write Cycle | $\overline{\text{RMC}}$  | Provides an indicator that the current bus cycle is part of an indivisible read-modify-write operation.                                                        |
| Address Strobe          | $\overline{\text{AS}}$   | Indicates that a valid address is on the bus.                                                                                                                  |
| Data Strobe             | $\overline{\text{DS}}$   | Indicates that valid data is to be placed on the data bus by an external device or has been placed on the data bus by the MC68030.                             |
| Data Buffer Enable      | $\overline{\text{DBEN}}$ | Provides an enable signal for external data buffers.                                                                                                           |

**Table 5-1. Signal Index (Sheet 2 of 2)**

| Signal Name                        | Mnemonic                                   | Function                                                                                                                                                                                                                          |
|------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Transfer and Size Acknowledge | $\overline{DSACK0}$<br>$\overline{DSACK1}$ | Bus response signals that indicate the requested data transfer operation is completed. In addition, these two lines indicate the size of the external bus port on a cycle-by-cycle basis and are used for asynchronous transfers. |
| Synchronous Termination            | $\overline{STERM}$                         | Bus response signal that indicates a port size of 32 bits and that data may be latched on the next falling clock edge.                                                                                                            |
| Cache Inhibit In                   | $\overline{CIIN}$                          | Prevents data from being loaded into the MC68030 instruction and data caches.                                                                                                                                                     |
| Cache Inhibit Out                  | $\overline{CIOUT}$                         | Reflects the CI bit in ATC entries or TTx register; indicates that external caches should ignore these accesses.                                                                                                                  |
| Cache Burst Request                | $\overline{CBREQ}$                         | Indicates a burst request for the instruction or data cache.                                                                                                                                                                      |
| Cache Burst Acknowledge            | $\overline{CBACK}$                         | Indicates that the accessed device can operate in burst mode.                                                                                                                                                                     |
| Interrupt Priority Level           | $\overline{IPL0-IPL2}$                     | Provides an encoded interrupt level to the processor.                                                                                                                                                                             |
| Interrupt Pending                  | $\overline{IPEND}$                         | Indicates that an interrupt is pending.                                                                                                                                                                                           |
| Autovector                         | $\overline{AVEC}$                          | Requests an autovector during an interrupt acknowledge cycle.                                                                                                                                                                     |
| Bus Request                        | $\overline{BR}$                            | Indicates that an external device requires bus mastership.                                                                                                                                                                        |
| Bus Grant                          | $\overline{BG}$                            | Indicates that an external device may assume bus mastership.                                                                                                                                                                      |
| Bus Grant Acknowledge              | $\overline{BGACK}$                         | Indicates that an external device has assumed bus mastership.                                                                                                                                                                     |
| Reset                              | $\overline{RESET}$                         | System reset.                                                                                                                                                                                                                     |
| Halt                               | $\overline{HALT}$                          | Indicates that the processor should suspend bus activity.                                                                                                                                                                         |
| Bus Error                          | $\overline{BERR}$                          | Indicates that an erroneous bus operation is being attempted.                                                                                                                                                                     |
| Cache Disable                      | $\overline{CDIS}$                          | Dynamically disables the on-chip cache to assist emulator support.                                                                                                                                                                |
| MMU Disable                        | $\overline{MMUDIS}$                        | Dynamically disables the translation mechanism of the MMU.                                                                                                                                                                        |
| Pipe Refill                        | $\overline{REFILL}$                        | Indicates when the MC68030 is beginning to fill pipeline.                                                                                                                                                                         |
| Microsequencer Status              | $\overline{STATUS}$                        | Indicates the state of the microsequencer.                                                                                                                                                                                        |
| Clock                              | CLK                                        | Clock input to the processor.                                                                                                                                                                                                     |
| Power Supply                       | V <sub>CC</sub>                            | Power supply.                                                                                                                                                                                                                     |
| Ground                             | GND                                        | Ground connection.                                                                                                                                                                                                                |

## 5.2 FUNCTION CODE SIGNALS (FC0–FC2)

These three-state outputs identify the address space of the current bus cycle. Table 4-1 shows the relationship of the function code signals to the privilege levels and the address spaces. Refer to **4.2 ADDRESS SPACE TYPES** for more information.

## 5.3 ADDRESS BUS (A0–A31)

These three-state outputs provide the address for the current bus cycle, except in the CPU address space. Refer to **4.2 ADDRESS SPACE TYPES** for more information on the CPU address space. A31 is the most significant address signal. Refer to **7.1.2 Address Bus** for information on the address bus and its relationship to bus operation.

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## 5.4 DATA BUS (D0–D31)

These three-state bidirectional signals provide the general-purpose data path between the MC68030 and all other devices. The data bus can transfer 8, 16, 24, or 32 bits of data per bus cycle. D31 is the most significant bit of the data bus. Refer to **7.1.4 Data Bus** for more information on the data bus and its relationship to bus operation.

## 5.5 TRANSFER SIZE SIGNALS (SIZ0, SIZ1)

These three-state outputs indicate the number of bytes remaining to be transferred for the current bus cycle. With A0, A1,  $\overline{\text{DSACK0}}$ ,  $\overline{\text{DSACK1}}$ , and  $\overline{\text{STERM}}$ , SIZ0 and SIZ1 define the number of bits transferred on the data bus. Refer to **7.2.1 Dynamic Bus Sizing** for more information on the size signals and their use in dynamic bus sizing.

## 5.6 BUS CONTROL SIGNALS

The following signals control synchronous bus transfer operations for the MC68030.

### 5.6.1 Operand Cycle Start ( $\overline{\text{OCS}}$ )

This output signal indicates the beginning of the first external bus cycle for an instruction prefetch or a data operand transfer.  $\overline{\text{OCS}}$  is not asserted for subsequent cycles that are performed due to dynamic bus sizing or operand misalignment. Refer to **7.1.1 Bus Control Signals** for information about the relationship of  $\overline{\text{OCS}}$  to bus operation.

### 5.6.2 External Cycle Start ( $\overline{\text{ECS}}$ )

This output signal indicates the beginning of a bus cycle of any type. Refer to **7.1.1 Bus Control Signals** for information about the relationship of  $\overline{\text{ECS}}$  to bus operation.

### 5.6.3 Read/Write ( $\text{R}/\overline{\text{W}}$ )

This three-state output signal defines the type of bus cycle. A high level indicates a read cycle; a low level indicates a write cycle. Refer to **7.1.1 Bus Control Signals** for information about the relationship of  $\text{R}/\overline{\text{W}}$  to bus operation.

### 5.6.4 Read-Modify-Write Cycle ( $\overline{\text{RMC}}$ )

This three-state output signal identifies the current bus cycle as part of an indivisible read-modify-write operation; it remains asserted during all bus cycles of the read-modify-write operation. Refer to **7.1.1 Bus Control Signals** for information about the relationship of  $\overline{\text{RMC}}$  to bus operation.

### 5.6.5 Address Strobe ( $\overline{\text{AS}}$ )

This three-state output indicates that a valid address is on the address bus. The function code, size, and read/write signals are also valid when  $\overline{\text{AS}}$  is asserted. Refer to **7.1.3 Address Strobe** for information about the relationship of  $\overline{\text{AS}}$  to bus operation.

## 5.6.6 Data Strobe ( $\overline{DS}$ )

During a read cycle, this three-state output indicates that an external device should place valid data on the data bus. During a write cycle, the data strobe indicates that the MC68030 has placed valid data on the bus. During two-clock synchronous write cycles, the MC68030 does not assert  $\overline{DS}$ . Refer to **7.1.5 Data Strobe** for more information about the relationship of  $\overline{DS}$  to bus operation.

## 5.6.7 Data Buffer Enable ( $\overline{DBEN}$ )

This output is an enable signal for external data buffers. This signal may not be required in all systems. The timing of this signal may preclude its use in a system that supports two-clock synchronous bus cycles. Refer to **7.1.6 Data Buffer Enable** for more information about the relationship of  $\overline{DBEN}$  to bus operation.

## 5.6.8 Data Transfer and Size Acknowledge ( $\overline{DSACK0}$ , $\overline{DSACK1}$ )

These inputs indicate the completion of a requested data transfer operation. In addition, they indicate the size of the external bus port at the completion of each cycle. These signals apply only to asynchronous bus cycles. Refer to **7.1.7 Bus Cycle Termination Signals** for more information on these signals and their relationship to dynamic bus sizing.

## 5.6.9 Synchronous Termination ( $\overline{STERM}$ )

This input is a bus handshake signal indicating that the addressed port size is 32 bits and that data is to be latched on the next falling clock edge for a read cycle. This signal applies only to synchronous operation. Refer to **7.1.7 Bus Cycle Termination Signals** for more information about the relationship of  $\overline{STERM}$  to bus operation.



## 5.7 CACHE CONTROL SIGNALS

The following signals relate to the on-chip caches.

### 5.7.1 Cache Inhibit Input ( $\overline{\text{CIIN}}$ )

This input signal prevents data from being loaded into the MC68030 instruction and data caches. It is a synchronous input signal and is interpreted on a bus-cycle-by-bus-cycle basis.  $\overline{\text{CIIN}}$  is ignored during all write cycles. Refer to **6.1 ON-CHIP CACHE ORGANIZATION AND OPERATION** for information on the relationship of  $\overline{\text{CIIN}}$  to the on-chip caches.

### 5.7.2 Cache Inhibit Output ( $\overline{\text{CIOUT}}$ )

This three-state output signal reflects the state of the CI bit in the address translation cache entry for the referenced logical address, indicating that an external cache should ignore the bus transfer. When the referenced logical address is within an area specified for transparent translation, the CI bit of the appropriate transparent translation register controls the state of  $\overline{\text{CIOUT}}$ . Refer to **SECTION 9 MEMORY MANAGEMENT UNIT** for more information about the address translation cache and transparent translation. Also, refer to **SECTION 6 ON-CHIP CACHE MEMORIES** for the effect of  $\overline{\text{CIOUT}}$  on the internal caches.

### 5.7.3 Cache Burst Request ( $\overline{\text{CBREQ}}$ )

This three-state output signal requests a burst mode operation to fill a line in the instruction or data cache. Refer to **6.1.3 Cache Filling** for filling information and **7.3.7 Burst Operation Cycles** for bus cycle information pertaining to burst mode operations.

### 5.7.4 Cache Burst Acknowledge ( $\overline{\text{CBACK}}$ )

This input signal indicates that the accessed device can operate in the burst mode and can supply at least one more long word for the instruction or data cache. Refer to **7.3.7 Burst Operation Cycles** for information about burst mode operation.

## 5.8 INTERRUPT CONTROL SIGNALS

The following signals are the interrupt control signals for the MC68030.

### 5.8.1 Interrupt Priority Level Signals

These input signals provide an indication of an interrupt condition and the encoding of the interrupt level from a peripheral or external prioritizing circuitry.  $\overline{IPL2}$  is the most significant bit of the level number. For example, since the  $\overline{IPLn}$  signals are active low,  $\overline{IPL0}$ – $\overline{IPL2}$  equal to \$5 corresponds to an interrupt request at interrupt level 2. Refer to **8.1.9 Interrupt Exceptions** for information on MC68030 interrupts.

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### 5.8.2 Interrupt Pending ( $\overline{IPEND}$ )

This output signal indicates that an interrupt request has been recognized internally and exceeds the current interrupt priority mask in the status register (SR). This output is for use by external devices (coprocessors and other bus masters, for example) to predict processor operation on the following instruction boundaries. Refer to **8.1.9 Interrupt Exceptions** for interrupt information. Also, refer to **7.4.1 Interrupt Acknowledge Bus Cycles** for bus information related to interrupts.

### 5.8.3 Autovector ( $\overline{AVEC}$ )

This input signal indicates that the MC68030 should generate an automatic vector during an interrupt acknowledge cycle. Refer to **7.4.1.2 AUTOVECTOR INTERRUPT ACKNOWLEDGE CYCLE** for more information about automatic vectors.

## 5.9 BUS ARBITRATION CONTROL SIGNALS

The following signals are the three bus arbitration control signals used to determine which device in a system is the bus master.

### 5.9.1 Bus Request ( $\overline{BR}$ )

This input signal indicates that an external device needs to become the bus master. This is typically a “wire-ORed” input (but does not need to be constructed from open-collector devices). Refer to **7.7 BUS ARBITRATION** for more information.

## 5.9.2 Bus Grant ( $\overline{\text{BG}}$ )

This output indicates that the MC68030 will release ownership of the bus master when the current processor bus cycle completes. Refer to **7.7.2 Bus Grant** for more information.

## 5.9.3 Bus Grant Acknowledge ( $\overline{\text{BGACK}}$ )

This input indicates that an external device has become the bus master. Refer to **7.7.3 Bus Grant Acknowledge** for more information.

# 5.10 BUS EXCEPTION CONTROL SIGNALS

The following signals are the bus exception control signals for the MC68030.

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## 5.10.1 Reset ( $\overline{\text{RESET}}$ )

This bidirectional open-drain signal is used to initiate a system reset. An external reset signal resets the MC68030 as well as all external devices. A reset signal from the processor (asserted as part of the RESET instruction) resets external devices only; the internal state of the processor is not altered. Refer to **7.8 RESET OPERATION** for a description of reset bus operation and **8.1.1 Reset Exception** for information about the reset exception.

## 5.10.2 Halt ( $\overline{\text{HALT}}$ )

The halt signal indicates that the processor should suspend bus activity or, when used with  $\overline{\text{BERR}}$ , that the processor should retry the current cycle. Refer to **7.5 BUS EXCEPTION CONTROL CYCLES** for a description of the effects of  $\overline{\text{HALT}}$  on bus operations.

## 5.10.3 Bus Error ( $\overline{\text{BERR}}$ )

The bus error signal indicates that an invalid bus operation is being attempted or, when used with  $\overline{\text{HALT}}$ , that the processor should retry the current cycle. Refer to **7.5 BUS EXCEPTION CONTROL CYCLES** for a description of the effects of  $\overline{\text{BERR}}$  on bus operations.

## 5.11.1 EMULATOR SUPPORT SIGNALS

The following signals support emulation by providing a means for an emulator to disable the on-chip caches and memory management unit and by supplying internal status information to an emulator. Refer to **SECTION 12 APPLICATIONS INFORMATION** for more detailed information on emulation support.

### 5.11.1 Cache Disable ( $\overline{\text{CDIS}}$ )

The cache disable signal dynamically disables the on-chip caches to assist emulator support. Refer to **6.1 ON-CHIP CACHE ORGANIZATION AND OPERATION** for information about the caches; refer to **SECTION 12 APPLICATIONS INFORMATION** for a description of the use of this signal by an emulator.  $\overline{\text{CDIS}}$  does not flush the data and instruction caches; entries remain unaltered and become available again when  $\overline{\text{CDIS}}$  is negated.

### 5.11.2 MMU Disable ( $\overline{\text{MMUDIS}}$ )

The MMU disable signal dynamically disables the translation of addresses by the MMU. Refer to **9.4 ADDRESS TRANSLATION CACHE** for a description of address translation; refer to **SECTION 12 APPLICATIONS INFORMATION** for a description of the use of this signal by an emulator. The assertion of  $\overline{\text{MMUDIS}}$  does not flush the address translation cache (ATC); ATC entries become available again when  $\overline{\text{MMUDIS}}$  is negated.

### 5.11.3 Pipeline Refill ( $\overline{\text{REFILL}}$ )

The pipeline refill signal indicates that the MC68030 is beginning to refill the internal instruction pipeline. Refer to **SECTION 12 APPLICATIONS INFORMATION** for a description of the use of this signal by an emulator.

### 5.11.4 Internal Microsequencer Status ( $\overline{\text{STATUS}}$ )

The microsequencer status signal indicates the state of the internal microsequencer. The varying number of clocks for which this signal is asserted indicates instruction boundaries, pending exceptions, and the halted condition. Refer to **SECTION 12 APPLICATIONS INFORMATION** for a description of the use of this signal by an emulator.

## 5.12 CLOCK (CLK)

The clock signal is the clock input to the MC68030. It is a TTL-compatible signal. Refer to **SECTION 12 APPLICATIONS INFORMATION** for suggestions on clock generation.

## 5.13 POWER SUPPLY CONNECTIONS

The MC68030 requires connection to a  $V_{CC}$  power supply, positive with respect to ground. The  $V_{CC}$  connections are grouped to supply adequate current for the various sections of the processor. The ground connections are similarly grouped. **SECTION 14 ORDERING INFORMATION AND MECHANICAL DATA** describes the groupings of  $V_{CC}$  and ground connections, and **SECTION 12 APPLICATIONS INFORMATION** describes a typical power supply interface.

## 5.14 SIGNAL SUMMARY

Table 5-2 provides a summary of the electrical characteristics of the signals discussed in this section.

**Table 5-2. Signal Summary**

| Signal Function                    | Signal Name                       | Input/Output | Active State | Three-State |
|------------------------------------|-----------------------------------|--------------|--------------|-------------|
| Function Codes                     | FC0–FC2                           | Output       | High         | Yes         |
| Address Bus                        | A0–A31                            | Output       | High         | Yes         |
| Data Bus                           | D0–D31                            | Input/Output | High         | Yes         |
| Transfer Size                      | SIZ0/SIZ1                         | Output       | High         | Yes         |
| Operand Cycle Start                | $\overline{\text{OCS}}$           | Output       | Low          | No          |
| External Cycle Start               | $\overline{\text{ECS}}$           | Output       | Low          | No          |
| Read/Write                         | R/W                               | Output       | High/Low     | Yes         |
| Read-Modify-Write Cycle            | $\overline{\text{RMC}}$           | Output       | Low          | Yes         |
| Address Strobe                     | $\overline{\text{AS}}$            | Output       | Low          | Yes         |
| Data Strobe                        | $\overline{\text{DS}}$            | Output       | Low          | Yes         |
| Data Buffer Enable                 | $\overline{\text{DBEN}}$          | Output       | Low          | Yes         |
| Data Transfer and Size Acknowledge | $\overline{\text{DSACK0/DSACK1}}$ | Input        | Low          | —           |
| Synchronous Termination            | $\overline{\text{STERM}}$         | Input        | Low          | —           |
| Cache Inhibit In                   | $\overline{\text{CIIN}}$          | Input        | Low          | —           |
| Cache Inhibit Out                  | $\overline{\text{CIOUT}}$         | Output       | Low          | Yes         |
| Cache Burst Request                | $\overline{\text{CBREQ}}$         | Output       | Low          | Yes         |
| Cache Burst Acknowledge            | $\overline{\text{CBACK}}$         | Input        | Low          | —           |
| Interrupt Priority Level           | $\overline{\text{IPL0–IPL2}}$     | Input        | Low          | —           |
| Interrupt Pending                  | $\overline{\text{IPEND}}$         | Output       | Low          | No          |
| Autovector                         | $\overline{\text{AVEC}}$          | Input        | Low          | —           |
| Bus Request                        | $\overline{\text{BR}}$            | Input        | Low          | —           |
| Bus Grant                          | $\overline{\text{BG}}$            | Output       | Low          | No          |
| Bus Grant Acknowledge              | $\overline{\text{BGACK}}$         | Input        | Low          | —           |
| Reset                              | $\overline{\text{RESET}}$         | Input/Output | Low          | No          |
| Halt                               | $\overline{\text{HALT}}$          | Input        | Low          | —           |
| Bus Error                          | $\overline{\text{BERR}}$          | Input        | Low          | —           |
| Cache Disable                      | $\overline{\text{CDIS}}$          | Input        | Low          | —           |
| MMU Disable                        | $\overline{\text{MMUDIS}}$        | Input        | Low          | —           |
| Pipeline Refill                    | $\overline{\text{REFILL}}$        | Output       | Low          | No          |
| Microsequencer Status              | $\overline{\text{STATUS}}$        | Output       | Low          | No          |
| Clock                              | CLK                               | Input        | —            | —           |
| Power Supply                       | VCC                               | Input        | —            | —           |
| Ground                             | GND                               | Input        | —            | —           |